

# PS-F03 | Autonomous Parametric Climate Insurance & Instant Settlement Engine

*By the time a traditional claim is adjudicated, the flood has already pushed the family into debt. A trigger-based payout doesn't have to wait for that.*

## 1. PROBLEM CONTEXT

Smallholder farmers, gig couriers, and informal workers in climate-vulnerable areas face catastrophic financial disruption from extreme weather — flash floods, heatwaves, and droughts. India ranks sixth globally in climate vulnerability and has recorded over 400 extreme weather events in the past three decades. Traditional indemnity-based insurance, including schemes like PMFBY, depends on lengthy post-event loss assessment — claim disputes, delayed payouts, and coverage gaps are common, well-documented grievances — leaving exactly the households least able to absorb a shock without emergency liquidity when they need it most. Parametric (index-based) insurance is emerging as a real, government-and-industry-backed alternative: Nagaland received a US\$119,000 payout in 2024 after a rainfall-trigger policy fired; SEWA and Global Parametrics have run gender-smart parametric pilots for smallholder women farmers across Rajasthan, Gujarat, and Maharashtra; and India's Ministry of Finance, NDMA, and GIC Re are reportedly exploring a nationwide parametric disaster-insurance scheme, with the market projected to grow ~11.3% annually through 2028. What remains an open engineering problem is the settlement infrastructure to make payouts genuinely instant and zero-touch, not merely faster than indemnity insurance. This challenge is written with a specific target architecture, at the organizers' request, rather than the fully solution-neutral style used elsewhere in this track — teams should treat the architecture below as the intended shape of the system, exercising their own judgement on stack and implementation details within it.

## 2. THE CORE CHALLENGE

Architect a fully automated, zero-touch parametric insurance engine that continuously streams weather telemetry, deterministically adjudicates policies against pre-agreed trigger conditions, and executes sub-minute payouts directly to digital wallets without requiring a manual claim.

## 3. TARGET STAKEHOLDERS / TECHNOLOGY OPPORTUNITY

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| <ul style="list-style-type: none"><li>• Smallholder farmers, gig couriers, and informal workers in climate-vulnerable regions</li><li>• Insurers, reinsurers, and microfinance institutions underwriting parametric products</li><li>• NDMA, IRDAI, and Ministry of Finance as the emerging regulatory/disaster-finance context</li><li>• Weather-data providers (satellite earth-observation operators, IoT weather-station networks, municipal agencies)</li></ul> | <ul style="list-style-type: none"><li>• Multi-oracle ingestion: an event-driven pipeline ingesting, deduplicating, and validating telemetry from independent sources (satellite, IoT stations, municipal APIs)</li><li>• Fault-tolerant adjudication: a consensus layer rejecting spoofed or corrupted telemetry before it triggers a payout</li><li>• Automated liquidity management: an algorithmic pool keeping capital dynamically rebalanced across high-risk micro-regions</li><li>• Dual-rail payout: an idempotent disbursement engine using instant fiat rails (e.g. UPI) or stablecoin rails if local infrastructure is degraded</li></ul> |
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## 4. DESIRED OUTCOME

- Payout latency reduced from months to sub-minute, trigger-based settlement
- Removal of manual claims assessment as a point of delay, dispute, or exclusion
- Continuity of capital availability across simultaneously affected micro-regions during a disaster event

## 5. SOLUTION SCOPE

- **Minimum:** an ingestion pipeline that reads simulated/public weather telemetry, evaluates it against a defined policy trigger, and executes a simulated payout to a synthetic wallet
- **Core workflow:** multi-source telemetry ingestion → validation/consensus → deterministic trigger evaluation → idempotent payout execution → audit trail
- **Optional extension:** Byzantine fault-tolerant multi-oracle consensus, algorithmic liquidity rebalancing across micro-regions, or dual-rail (fiat/stablecoin) payout routing

## 6. DATA & RESOURCE AVAILABILITY

Permitted inputs: Public weather/earth-observation data (IMD data, open satellite precipitation/temperature datasets, open IoT station feeds); NPCI's public UPI documentation as a fiat-rail reference; documented trigger structures from existing Indian parametric pilots; synthetic wallet, policy, and payout data built by the team — no real telemetry contracts, production payment rails, or real money will be provided or used.

**Organizer-provided resources:** None. No proprietary datasets, private resources, or special infrastructure will be provided by the organizers.

## 7. END-TO-END DEMONSTRATION

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**Input:** continuous weather telemetry from multiple simulated/public sources for a defined micro-region → **Processing:** ingestion, deduplication, and cross-source validation of the incoming telemetry → **Decision:** deterministic evaluation of the validated data against a pre-agreed policy trigger → **Action:** an idempotent payout is executed to a synthetic digital wallet the moment the trigger fires → **Outcome:** a policyholder receives settlement within a sub-minute window of the triggering event, with no manual claim filed

## 8. SUCCESS METRICS / RESPONSIBLE FINTECH & SAFETY BOUNDARIES

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| <ul style="list-style-type: none"><li>• End-to-end latency from trigger condition met to payout executed</li><li>• Telemetry validation/consensus accuracy (rejection rate of spoofed or corrupted synthetic data)</li><li>• Payout idempotency — no duplicate or missed disbursement under retried/failed-delivery conditions</li><li>• False-trigger and missed-trigger rate against a labelled synthetic weather-event test set (basis risk)</li></ul> | <ul style="list-style-type: none"><li>• No real money, real bank/wallet accounts, or real payment-rail integration (UPI, FedNow, stablecoin) may be used — all disbursement must be simulated against synthetic wallets</li><li>• No real telemetry-provider contracts or production satellite/IoT data feeds are required or should be sought — public or synthetic telemetry only</li><li>• The prototype must not claim regulatory approval, insurer licensing, or production-grade security guarantees it has not actually demonstrated</li><li>• Basis risk (a real loss occurring without the parameter threshold being crossed) must be explicitly acknowledged as a limitation, not designed around silently</li></ul> |
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## 9. EVALUATION FOCUS

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Problem understanding • Technical depth (oracle consensus, idempotent settlement) • Latency and reliability of the pipeline • Resilience to corrupted/adversarial telemetry • Responsible framing of guarantees made • Quality of demonstration

## 10. SUGGESTED DEMO SCENARIO

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*A flash flood hits a district where a smallholder farmer holds a rainfall-trigger parametric policy. Using the prototype, simulated telemetry from multiple independent weather sources streams in and is cross-validated; once the rainfall threshold is confirmed as crossed by consensus among the sources, the system deterministically fires the policy and executes a payout to the farmer's synthetic wallet within the same minute — with no claim form, adjuster visit, or waiting period.*

## 11. EXPECTED DELIVERABLE

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A working prototype demonstrating the full telemetry-to-payout pipeline on public/synthetic weather data and synthetic wallets; a technical architecture diagram covering ingestion, consensus, and settlement; measured latency, consensus accuracy, and idempotency under simulated failure/retry conditions; and an honest discussion of basis risk and what production deployment would require beyond the prototype.

## 12. EVIDENCE & REFERENCES

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- India explores nationwide insurance for climate disaster risks, Insurance Business — <https://www.insurancebusinessmag.com/asia/news/catastrophe/india-explores-nationwide-insurance-for-climate-disaster-risks-552191.aspx>
  - Parametric insurance: Protecting businesses and livelihoods, WEF — <https://www.weforum.org/stories/2025/03/the-heatwave-that-sparked-a-new-era-for-women-workers-in-india/>
  - Parametric Insurance in India: The Future of Climate-Resilient Agriculture — <https://www.insurancesamadhan.com/blog/parametric-insurance-in-india-the-future-of-climate-resilient-agriculture/>
  - Parametric insurance: India's new climate safety net, Policy Circle — <https://www.policycircle.org/policy/parametric-insurance-climate-safety/>
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